

THEOREM R: SHARP TOPOLOGY-INDEPENDENT STABILITY OF REAL PROJECTED MULTILINEAR TREES

REVIEW NOTE

ABSTRACT. A projected multilinear tree composes independently chosen multilinear maps between finite-dimensional real Hilbert spaces along a rooted tree and projects orthogonally after every internal operation. Under a uniform operator-norm budget M and a closure budget $\rho = \eta M$ on the part of each law that leaves the retained subspace on projected inputs, let $C_T^P(\eta)$ be the sharp constant for the projected output error. We prove that

$$C_T^P(\eta) = \frac{1}{\eta} \max_{0 \leq \theta \leq \arcsin \eta} |1 - (\cos \theta e^{i\theta})^{k-1}|,$$

where k is the number of internal nodes of T . The constant depends on the number of internal nodes and on the leakage, but not on depth or topology. The upper bound rests on a Löwner-order state invariant and a tensor angle inequality for projective norms; the lower bound is an explicit construction valid for every tree.

INTRODUCTION

The problem. A computation that alternates multilinear operations with orthogonal reductions is modelled by a rooted tree. Each internal node applies a multilinear law to its children's values, and a projector keeps only a retained subspace. Examples include truncated tensor-network contractions, Galerkin projections of polynomial nonlinearities, and reduced models of recursive compositions. The recursively reduced value R_r is compared with the ambient value F_r computed without any reduction. The natural question is extremal: if every law has norm at most M and leaks at most ρ out of its retained subspace on already-reduced inputs, how large can the projected error $\|P_r F_r - R_r\|$ be?

What was known. Telescoping over arguments gives the universal bound

$$(1) \quad \|P_r F_r - R_r\| \leq (k-1) \rho M^{k-1} \prod_{\ell} \|z_{\ell}\|.$$

Indeed, by induction $\|F_v - R_v\| \leq k_v \rho M^{k_v-1} L_v$ at every node. The difference $\mu_v(F_c) - \mu_v(R_c)$ telescopes into one term per child, and the local defect $Q_v \mu_v(R_c)$ contributes one more ρ ; at the root this defect is annihilated by P_r . The bound (1) is asymptotically sharp as $\eta = \rho/M \rightarrow 0$. For $k = 2$ it is exact ($C_2^P \equiv 1$). For $k = 3$ the exact constant is $W_3(\eta) = \sqrt{4 - 3\eta^2}$ for $\eta \leq \sqrt{2/3}$ and $2/(\sqrt{3}\eta)$ above, for both three-node skeletons. This shows that orthogonality forbids simultaneous saturation of all local errors at positive leakage.

What is proved here. For every tree the sharp constant is a single explicit function of the number k of internal nodes and the leakage η (Theorem R). The whole family of trees collapses to the one-parameter sequence $C_1, C_2, C_3, \dots$. Section 10 records the resulting closed forms, the second-order coefficient, and the behaviour as $k \rightarrow \infty$.

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The idea. The state of a node is the 2×2 Gram matrix of its ambient and reduced values. States are ordered by Löwner domination against the Gram matrix of a unit vector and a vector of norm ρ at an angle at most χ (order O2). Two facts make this order propagate through the tree:

- a projection step multiplies the chain label $\rho e^{i\chi}$ by $w(\theta) = \cos \theta e^{i\theta}$, because isometric dilations preserve angles and the closure budget;
- a multilinear node multiplies the labels of its children, by a tensor angle inequality for projective norms.

The labels therefore compose as complex numbers, and the error is at most $|1 - \prod w(\theta_v)|$ over the $k - 1$ non-root internal nodes. A concavity argument then shows that equal angles are extremal. Conversely, multiplication in $\mathbb{R}^2 \cong \mathbb{C}$ realizes every such product exactly, for every tree.

Relation to the literature. The triangle inequality for angles under nonexpansive maps also underlies the scaled relative graph calculus [1]. Isometric dilations of contractions are classical [3], and projective tensor norms are standard [2]. We are not aware of the tensor angle inequality (Theorem 3.1) or of Theorem R in the literature, but a systematic comparison has not been carried out, and no novelty is claimed.

How to read this note. The argument splits into two independent halves:

upper bound: Definitions $\rightarrow$ O2 $\rightarrow$ Tensor angle lemma $\rightarrow$ Lemma 3
 $\rightarrow$ Multiplicative envelope $\rightarrow$ Diagonal Lemma,
 lower bound: Universal sharp witness.

Sections 3 and 6 use no tree vocabulary and can be read on their own. The step we regard as the one most in need of independent scrutiny is the passage *tensor angle inequality* $\Rightarrow$ *preservation of the order O2 at a multilinear node* (Theorem 3.1 and Lemma 4.2). It is also available as a separate short note.

Throughout, k denotes the *number of internal nodes*, not the depth. A chain with k internal nodes has depth k ; a star with k internal nodes has depth 2.

1. DEFINITIONS

1.1. The class **PMT-A**.

- (A1) T is a finite rooted tree with internal nodes V_{int} , $k := |V_{\text{int}}| \geq 1$, root r , and leaves L . Each internal node v has $m_v \geq 1$ ordered children, some of which may be leaves and some internal.
- (A2) Every node w carries a finite-dimensional real Hilbert space H_w of arbitrary dimension (dimension 2 is allowed).
- (A3) Every internal v carries an orthogonal projector P_v of arbitrary rank. Rank 0, rank 1 and $P_v = I$ are all allowed, also at the root. Write $Q_v := I - P_v$.
- (A4) Every internal v carries a multilinear map μ_v from the product of its children's spaces to H_v , with $\|\mu_v\| := \sup\{\|\mu_v(x_1, \dots)\| : \|x_i\| \leq 1\} \leq M$. Laws at different nodes are chosen independently.
- (A5) Leaves carry nonzero data z_ℓ . Set $F_\ell = R_\ell = z_\ell$, and for internal v

$$F_v = \mu_v(F_{\text{children}}), \quad R_v = P_v \mu_v(R_{\text{children}}).$$

- (A6) *Closure.* For every internal v , $\|Q_v \mu_v(x_1, \dots)\| \leq \rho \prod \|x_i\|$ whenever each internal-child slot receives a vector in that child's $\text{Ran } P$. Leaf slots receive arbitrary vectors. Here $0 < \rho \leq M$ and $\eta := \rho/M \in (0, 1]$; the value $\eta = 0$ is not part of the class.

(A7) The projected error is $E_T^P := \|P_r F_r - R_r\|$, and

$$C_T^P(\eta) := \sup \frac{E_T^P}{\rho M^{k-1} \prod_\ell \|z_\ell\|},$$

the supremum over all admissible realizations, dimensions and ranks.

1.2. Normalization and trajectory closure.

- (N1) *Scaling.* The ratio is invariant under $\mu_v \mapsto \lambda \mu_v$ and $z_\ell \mapsto \nu_\ell z_\ell$. We take $M = 1$, $\rho = \eta$ and unit leaves.
- (N2) *Leaf freezing.* Fix leaf arguments at the leaf data. At each internal v with internal children $c_1, \dots, c_m$ ($m \geq 0$) this gives a multilinear map on the internal slots with norm ≤ 1 , the same F and R , and

$$(TC) \quad \|Q_v \mu_v(R_{c_1}, \dots, R_{c_m})\| \leq \eta \prod_i \|R_{c_i}\|,$$

because $R_{c_i} \in \text{Ran } P_{c_i}$. For $m = 0$, μ_v is a vector f_v with $\|f_v\| \leq 1$ and $\|Q_v f_v\| \leq \eta$.

The upper bound uses only norms ≤ 1 , orthogonality of each P_v , and (TC). The witness satisfies the full closure (A6).

1.3. The theorem. Let

$$w(\theta) := \cos \theta e^{i\theta} = \frac{1}{2}(1 + e^{2i\theta}), \quad C_k(\eta) := \frac{1}{\eta} \max_{0 \leq \theta \leq \arcsin \eta} |1 - w(\theta)^{k-1}|.$$

Theorem R. For every finite rooted tree T in real PMT-A with k internal nodes and every $\eta \in (0, 1]$,

$$C_T^P(\eta) = C_k(\eta).$$

Remark. At $k = 2, 3$ this recovers $C_2^P = 1$ and the known two-regime constant $W_3(\eta) = \sqrt{4 - 3\eta^2}$ for $\eta \leq \sqrt{2/3}$, $2/(\sqrt{3}\eta)$ above. As $\eta \downarrow 0$, $C_k(\eta) = (k-1) - \frac{(k-1)(k-2)(k+6)}{24}\eta^2 + O(\eta^4)$.

2. THE ORDER O2

For $\rho \geq 0$ and $\psi \in [0, \pi]$ put

$$K(\rho, \psi) := \begin{pmatrix} 1 & \rho \cos \psi \\ \rho \cos \psi & \rho^2 \end{pmatrix},$$

the Gram matrix of a unit vector and a vector of norm ρ at angle ψ . Angles are $\angle(a, b) = \arccos(\langle a, b \rangle / (\|a\| \|b\|)) \in [0, \pi]$; this is a metric on the unit sphere. $\preceq$ denotes the Löwner order.

Definition 2.1 (O2). A pair (F, R) is *dominated by the label* (ρ, χ) , with $\rho, \chi \geq 0$, if

$$\text{Gram}(F, R) \preceq K(\rho, \psi') \quad \text{for some } \psi' \in [0, \min(\chi, \pi)].$$

The label χ is a cumulative phase bookkeeping parameter. It is not reconstructed from the Gram matrix.

Lemma 2.2 (Realization). Assume $\text{Gram}(F, R) \preceq K(\rho, \psi)$ with $\rho \geq 0$, $\psi \in [0, \pi]$. Let X be a 2-dimensional real space with unit vectors $\hat{F}, \hat{R}$ at angle ψ . Then there is $B : X \rightarrow H$ with $\|B\| \leq 1$, $B\hat{F} = F$ and $B(\rho\hat{R}) = R$. In particular $\|F\| \leq 1$ and $\|R\| \leq \rho$.

Proof. For real α, β we have $\|\alpha F + \beta R\|^2 \leq \|\alpha \hat{F} + \beta \rho \hat{R}\|^2$. So $B(\alpha \hat{F} + \beta \rho \hat{R}) := \alpha F + \beta R$ is well defined and contractive on the span. Extend it by 0 on the orthogonal complement. This covers $\rho = 0$ (forcing $R = 0$) and $\psi \in \{0, \pi\}$ (forcing $R = \pm \rho F$). $\square$

Lemma 2.3 ((N−) implies O2). *Let $S \in [0, \pi]$ and let f, g satisfy*

$$(N-) \quad \|af + bg\| \leq |ae^{iS} + b| \quad \text{for all real } a, b \text{ with } ab \leq 0.$$

Then $\text{Gram}(f, g) \preceq K(1, \psi)$ for some $\psi \in [0, S]$.

Proof. Let $G = \text{Gram}(f, g)$.

- (i) The choices $b = 0$ and $a = 0$ give $G_{11}, G_{22} \leq 1$. Put $\mu := \sqrt{(1 - G_{11})(1 - G_{22})}$.
- (ii) (N−) reads $a^2(1 - G_{11}) + b^2(1 - G_{22}) - 2|ab|(\cos S - G_{12}) \geq 0$. Minimizing over the ratio $|a| : |b|$, this holds for all a, b if and only if $G_{12} + \mu \geq \cos S$.
- (iii) Since its diagonal is nonnegative, $K(1, \psi) - G \succeq 0$ if and only if $\cos \psi \in [G_{12} - \mu, G_{12} + \mu]$.
- (iv) This interval meets $[\cos S, 1]$: $G_{12} - \mu \leq \sqrt{G_{11}G_{22}} \leq 1$, and $\cos S \leq G_{12} + \mu$ by (ii).

Taking $\cos \psi$ in the intersection gives $\psi \in [0, S]$. $\square$

Remark. (N−) bounds the alignment G_{12} only from below. Excess alignment is absorbed by a dilation angle $\psi < S$. This is the precise sense in which extra alignment cannot help the adversary.

3. THE TENSOR ANGLE LEMMA

For real Hilbert spaces $H_1, \dots, H_m$, $\|\cdot\|_\pi$ denotes the projective tensor norm on $H_1 \otimes \dots \otimes H_m$. It is the dual of the multilinear operator norm: $\|\mathbf{T}\|_\pi = \sup W(\mathbf{T})$ over m -linear forms W with $|W(x_1, \dots, x_m)| \leq \prod \|x_i\|$.

Theorem 3.1 (Tensor angle inequality). *Let $u_i, v_i \in H_i$ be unit vectors, $\psi_i := \angle(u_i, v_i)$ and $S := \sum_{i=1}^m \psi_i$. For all real a, b with $ab \leq 0$,*

$$\left\| a \bigotimes_{i=1}^m u_i + b \bigotimes_{i=1}^m v_i \right\|_\pi \leq |ae^{i \min(S, \pi)} + b|.$$

Proof. Reductions. We may take $a = 1$, $b = -t$ with $t \geq 0$. If $S \geq \pi$, the right side is $1 + t$ and the claim is the triangle inequality, so assume $S < \pi$. Factors of dimension one may be embedded isometrically into $\mathbb{R}^2$, since projective norms of Hilbert factors are preserved under isometric embeddings.

Induction on m . For $m = 1$ the claim is the identity $\|u - tv\|^2 = 1 + t^2 - 2t \cos \psi_1$. Let $m \geq 2$.

- (1) Put $X = \bigotimes_{i \geq 2} u_i$, $Y = \bigotimes_{i \geq 2} v_i$ and $S' = S - \psi_1$. By associativity of $\otimes_\pi$ and duality, the left side equals $\sup_\beta [\beta(u_1, X) - t\beta(v_1, Y)]$ over bilinear forms β on $H_1 \times (\bigotimes_{i \geq 2} H_i)$ with $|\beta(z, W)| \leq \|z\| \|W\|_\pi$.
- (2) Fix a 2-plane $\Pi \ni u_1, v_1$. By the Riesz representation, $\beta(z, W) = \langle z, \Lambda W \rangle$ for $z \in \Pi$, with $\|\Lambda W\| \leq \|W\|_\pi$.
- (3) Put $p = \Lambda X$ and $q = \Lambda Y$. The induction hypothesis gives $\|\alpha p + \beta' q\| \leq |\alpha e^{iS'} + \beta'|$ for $\alpha \beta' \leq 0$.
- (4) Lemmas 2.3 and 2.2 give unit vectors $\hat{p}, \hat{q}$ at an angle $\tilde{\psi} \leq S'$ and a contraction C with $C\hat{p} = p$, $C\hat{q} = q$.
- (5) Let Rot be the rotation of Π taking v_1 to u_1 . Then $\beta(u_1, X) - t\beta(v_1, Y) = \langle u_1, C\hat{p} - t\text{Rot } C\hat{q} \rangle \leq \|C\hat{p} - t\text{Rot } C\hat{q}\|$.
- (6) Let $V = (C, (I - C^*C)^{1/2})$, an isometry into $\Pi \oplus D$, and $\tilde{R} = \text{Rot} \oplus I$. For unit y , $\langle y, \tilde{R}y \rangle = \|y_\Pi\|^2 \cos \psi_1 + \|y_D\|^2 \geq \cos \psi_1$, so $\angle(y, \tilde{R}y) \leq \psi_1$.
- (7) The Π -component of $V\hat{p} - t\tilde{R}V\hat{q}$ is $C\hat{p} - t\text{Rot } C\hat{q}$. The vectors $V\hat{p}$ and $t\tilde{R}V\hat{q}$ have norms 1 and t , and the angle between them is at most $\tilde{\psi} + \psi_1 \leq S < \pi$.

Therefore the quantity in (5) is at most $|1 - te^{iS}|$. $\square$

Proposition 3.2 (Sharpness). *If $S \leq \pi$, equality holds in Theorem 3.1.*

Proof. Identify each $\Pi_i \cong \mathbb{C}$ with $v_i \mapsto 1$ and $u_i \mapsto e^{i\psi_i}$, and let π_i be the orthogonal projection onto Π_i . For $|\lambda| = 1$ the form $W(x) = \operatorname{Re}(\lambda \prod_i \pi_i x_i)$ has norm at most 1, and $W(\otimes u_i) - tW(\otimes v_i) = \operatorname{Re}(\lambda(e^{iS} - t))$. Choose λ . $\square$

4. LEMMA 3: O2 IS PRESERVED AT NON-ROOT INTERNAL NODES

Lemma 4.1 (Child substitution and closure normalization). *Let v be internal with internal children $c_1, \dots, c_m$, $m \geq 1$, let $\|\mu_v\| \leq 1$, let (TC) hold, and let each (F_{c_i}, R_{c_i}) be dominated by (ρ_i, χ_i) via $\psi_i \in [0, \min(\chi_i, \pi)]$. Let B_i be as in Lemma 2.2, with unit $\hat{F}_i, \hat{R}_i$ at angle ψ_i . Put $\mu'(x) := \mu_v(B_1 x_1, \dots, B_m x_m)$ and $\rho := \prod_i \rho_i$. Then:*

- (i) $\|\mu'\| \leq 1$;
- (ii) $F_v = \mu'(\hat{F}_1, \dots)$ and $\mu_v(R_{c_1}, \dots) = \rho \mu'(\hat{R}_1, \dots)$;
- (iii) if $\rho > 0$, then $\|Q_v \mu'(\hat{R}_1, \dots)\| \leq \eta$;
- (iv) if $\rho = 0$, then $\mu_v(R_{c_1}, \dots) = 0$.

Proof. (i) Composition with contractions. (ii) Multilinearity. (iii) By (ii) and (TC), $\|Q_v \mu'(\hat{R})\| = \|Q_v \mu_v(R_c)\|/\rho \leq \eta \prod \|R_{c_i}\|/\rho \leq \eta$, using $\|R_{c_i}\| \leq \rho_i$. (iv) Some $R_{c_i} = 0$. $\square$

Lemma 4.2 (Lemma 3). *Let $v \neq r$ be internal with $m \geq 0$ internal children satisfying the hypotheses of Lemma 4.1. Then there is $\theta_v \in [0, \arcsin \eta]$ such that (F_v, R_v) is dominated by*

$$(\rho_v, \chi_v) := \left(\cos \theta_v \prod_i \rho_i, \theta_v + \sum_i \chi_i \right)$$

(empty product 1, empty sum 0). When $\prod_i \rho_i > 0$ one can take

$$\sin \theta_v = \frac{\|Q_v \mu_v(R_{c_1}, \dots, R_{c_m})\|}{\prod_i \rho_i}.$$

Proof. Case $\prod_i \rho_i = 0$ (so $m \geq 1$). By Lemma 4.1(iv), $R_v = 0$, and $\|F_v\| \leq 1$. Hence $\operatorname{Gram}(F_v, 0) \preceq K(0, 0)$; take $\theta_v = 0$.

Case $\rho := \prod_i \rho_i > 0$.

- (1) *Raw output.* Put $f = \mu'(\hat{F})$ and $g = \mu'(\hat{R})$, so $F_v = f$ and $R_v = \rho P_v g$. For $m = 0$ put $f = g = f_v$, $\rho = 1$, $S = 0$; otherwise $S = \sum \psi_i$.
- (2) *(N-) for the raw output.* With $S' = \min(S, \pi)$, for $ab \leq 0$,

$$\|af + bg\| = \sup_{\|\omega\| \leq 1} \langle \omega, \mu'(a \otimes \hat{F} + b \otimes \hat{R}) \rangle \leq \|a \otimes \hat{F} + b \otimes \hat{R}\|_\pi \leq |ae^{iS'} + b|$$

by Theorem 3.1. For $m = 0$ this is immediate.

- (3) *Domination.* Lemmas 2.3 and 2.2 give unit $\hat{F}', \hat{E}'$ at an angle $\tilde{\psi} \leq S'$ and a contraction C with $C\hat{F}' = f$, $C\hat{E}' = g$.
- (4) *Dilation.* Let $V = (C, (I - C^*C)^{1/2})$ and $P' = P_v \oplus I$, and set $\tilde{F} = V\hat{F}'$, $\tilde{R} = \rho P'V\hat{E}'$. Their H_v -components are F_v and R_v , so $\operatorname{Gram}(F_v, R_v) \preceq \operatorname{Gram}(\tilde{F}, \tilde{R})$.
- (5) *Norms.* $\|\tilde{F}\| = 1$. Also $\|(I - P')V\hat{E}'\| = \|Q_v g\| \leq \eta$ by Lemma 4.1(iii) (by (N2) if $m = 0$). Put $\theta_v := \arcsin \|Q_v g\|$. Then $\|\tilde{R}\| = \rho \cos \theta_v$, and $\|Q_v g\| = \|Q_v \mu_v(R_c)\|/\rho$ gives the stated formula.
- (6) *Angle.* If $\tilde{R} = 0$, the pair is dominated by $K(0, 0)$. Otherwise $\angle(\tilde{F}, \tilde{R}) \leq \angle(V\hat{F}', V\hat{E}') + \angle(V\hat{E}', P'V\hat{E}') = \tilde{\psi} + \theta_v \leq \chi_v$, so $\operatorname{Gram}(\tilde{F}, \tilde{R}) = K(\rho_v, \angle(\tilde{F}, \tilde{R}))$.

Combining (4) and (6) proves the claim. $\square$

5. THE MULTIPLICATIVE ENVELOPE

Encode a label as $z = \rho e^{i\chi}$. Lemma 4.2 then reads $z_v = w(\theta_v) \prod_i z_{c_i}$.

Proposition 5.1 (Unrolling). *Applying Lemma 4.2 from the leaves upward, every non-root internal v has (F_v, R_v) dominated by $z_v = \prod_{u \in V_{\text{int}}(T_v)} w(\theta_u)$, where T_v is the subtree of v . Since the subtrees of the root's internal children are pairwise disjoint and exhaust $V_{\text{int}} \setminus \{r\}$,*

$$\prod_i z_{c_i} = \prod_{u \in V_{\text{int}} \setminus \{r\}} w(\theta_u),$$

a product with exactly $k - 1$ factors.

Remark. The count uses that T is a tree; for directed acyclic graphs this step fails as stated. O2 induces a commutative multiplicative envelope for tree states. We make no stronger algebraic claim.

Lemma 5.2 (Root reading). *Let the root have $m \geq 1$ internal children dominated by (ρ_i, χ_i) via angles ψ_i , and let $\|\mu_r\| \leq 1$. No closure is needed at the root. Then $E_T^P \leq |1 - \rho e^{i \min(S, \pi)}|$ with $\rho = \prod \rho_i$ and $S = \sum \psi_i$.*

Proof. $E_T^P = \|P_r(\mu_r(F_c) - \mu_r(R_c))\|$. Substitute the children (Lemma 4.1(i)–(ii)), scalarize with a unit $\omega \in \text{Ran } P_r$, and apply Theorem 3.1 with $a = 1$, $b = -\rho$. If $\rho = 0$ the bound is $1 \geq \|F_r\|$. $\square$

Corollary 5.3. *There are $\theta_u \in [0, \arcsin \eta]$, $u \in V_{\text{int}} \setminus \{r\}$, with*

$$E_T^P \leq |1 - R e^{i \min(\Theta, \pi)}|, \quad R = \prod_{u \neq r} \cos \theta_u, \quad \Theta = \sum_{u \neq r} \theta_u.$$

Proof. Lemma 5.2 with $S \leq \Theta$, since $|1 - R e^{i\varphi}|$ is nondecreasing in $\varphi \in [0, \pi]$ for $R \geq 0$. $\square$

6. THE DIAGONAL LEMMA

Throughout this section $n \geq 1$ and $\alpha \in (0, \pi/2]$.

Lemma 6.1 (Subadditivity of cosine products). *If $\theta_j \geq 0$ and $\sum_j \theta_j \leq \pi/2$, then $\prod_j \cos \theta_j \geq \cos(\sum_j \theta_j)$.*

Proof. Induction on n . All partial sums lie in $[0, \pi/2]$, so the sines and cosines involved are nonnegative. Then $\cos(a+b) = \cos a \cos b - \sin a \sin b \leq \cos a \cos b$, and multiplying the inductive inequality by $\cos \theta_n \geq 0$ preserves it. $\square$

Remark. The hypothesis $\sum \theta_j \leq \pi/2$ matters for large angle sums. With $n = 4$ and $\theta_j = \pi/2 - \varepsilon$ we get $\prod \cos \theta_j = \sin^4 \varepsilon \approx 0$ while $\cos \sum \theta_j \approx 1$. For $\sum \theta_j \in (\pi/2, \pi]$ the inequality holds trivially, because $\cos \sum \theta_j \leq 0$.

Lemma 6.2 (Jensen for log cos). *$\log \cos$ is strictly concave on $[0, \pi/2)$, since $(\log \cos)'' = -\sec^2 < 0$. Hence for $\theta_j \in [0, \pi/2)$ with $\Theta = \sum \theta_j$ we have $\prod_j \cos \theta_j \leq \cos^n(\Theta/n)$, with equality if and only if all θ_j are equal.*

Lemma 6.3 (Diagonal Lemma).

$$\max_{\theta \in [0, \alpha]^n} \left| 1 - \prod_{j=1}^n w(\theta_j) \right| = \max_{t \in [0, \alpha]} |1 - w(t)^n|.$$

Moreover, for every $\theta \in [0, \alpha]^n$, with $R = \prod \cos \theta_j$ and $\Theta = \sum \theta_j$,

$$|1 - R e^{i \min(\Theta, \pi)}| \leq \max_{t \in [0, \alpha]} |1 - w(t)^n|.$$

Proof. The inequality “ $\geq$ ” is the diagonal choice. For “ $\leq$ ”, write $\prod w(\theta_j) = Re^{i\Theta}$ and $\varphi := \min(\Theta, \pi)$, so that $|1 - Re^{i\varphi}|^2 = f_\varphi(R) := 1 - 2R\cos\varphi + R^2$. For $\Theta \leq \pi$ this is the value at θ . For $\Theta > \pi$ it dominates that value, because $|1 - Re^{i\Theta}| \leq 1 + R$.

Some $\theta_j = \pi/2$. Then $\alpha = \pi/2$, $R = 0$ and the value is 1, which the diagonal attains at $t = \pi/2$. Assume from now on that all $\theta_j < \pi/2$.

Case $\Theta \leq \pi$. First, $R \geq \cos\Theta$: by Lemma 6.1 if $\Theta \leq \pi/2$, and trivially otherwise. Next, $f'_\Theta(R) = 2(R - \cos\Theta) \geq 0$ on $[R, \cos^n(\Theta/n)]$, so Lemma 6.2 gives $f_\Theta(R) \leq f_\Theta(\cos^n(\Theta/n)) = |1 - w(\Theta/n)^n|^2$. Finally $\Theta/n \leq \alpha$.

Case $\Theta > \pi$. Then $n\alpha > \pi$, so $n \geq 3$ and $\pi/n < \alpha$. By Lemma 6.2 and the monotonicity of $\cos$,

$$|1 - Re^{i\varphi}| = 1 + R \leq 1 + \cos^n(\Theta/n) \leq 1 + \cos^n(\pi/n) = |1 - w(\pi/n)^n|,$$

since $w(\pi/n)^n = -\cos^n(\pi/n)$. $\square$

Remark. The split at $\Theta = \pi$ is necessary: for $\Theta > \pi$ the bound $R \geq \cos\Theta$, and with it the monotonicity in R , can fail.

7. UPPER BOUND

Theorem 7.1. $C_T^P(\eta) \leq C_k(\eta)$.

Proof. Normalize by (N1) and (N2). If $k = 1$, then $R_r = P_r F_r$ and $E_T^P = 0$. Otherwise Corollary 5.3 and the second part of Lemma 6.3, with $n = k-1$ and $\alpha = \arcsin \eta$, give $E_T^P \leq \eta C_k(\eta)$. $\square$

8. THE UNIVERSAL SHARP WITNESS

Theorem 8.1 (Universal sharp witness). *Let T be a finite rooted tree with k internal nodes, $\eta \in (0, 1]$, and choose $\theta_v \in [0, \arcsin \eta]$ for the non-root internal nodes. The realization below is PMT-A-admissible with $M = 1$, $\rho = \eta$ and unit leaves, and*

$$E_T^P = \left| 1 - \prod_{v \in V_{\text{int}} \setminus \{r\}} w(\theta_v) \right|.$$

Construction. All spaces are $\mathbb{R}^2 \cong \mathbb{C}$ with $e_0 = 1$, and all leaves carry e_0 . For an internal v , let I_v be its internal-child slots and L_v its leaf slots.

- At non-root v : $P_v(z) = \text{Re}(z)$ (rank one) and $\mu_v(x) = e^{i\theta_v} \prod_{j \in I_v} x_j \prod_{\ell \in L_v} \text{Re}(x_\ell)$ (complex product).
- At the root: $P_r = I$ and $\mu_r(x) = \prod_{j \in I_r} x_j \prod_{\ell \in L_r} \text{Re}(x_\ell)$.

Proof. Norms. Since $|\prod x_j| = \prod |x_j|$ and $|\text{Re}(x)| \leq |x|$, every law has norm 1.

Closure at non-root v . Take internal slots in $\text{Ran } P = \mathbb{R}$ and arbitrary leaf slots. Every factor is real, so the output is $e^{i\theta_v} t$ with t real and $|t| \leq \prod \|x\|$. Hence $\|Q_v \mu_v(x)\| = |\sin \theta_v| |t| \leq \eta \prod \|x\|$. At the root, $Q_r = 0$.

Evaluation. By induction, every non-root internal v satisfies $F_v = e^{i\Phi_v}$ and $R_v = A_v$, where $\Phi_v = \sum_{u \in V_{\text{int}}(T_v)} \theta_u$ and $A_v = \prod_{u \in V_{\text{int}}(T_v)} \cos \theta_u$. At a leaf-only node, $F_v = e^{i\theta_v}$ and $R_v = \text{Re}(e^{i\theta_v}) = \cos \theta_v$. In the inductive step, $R_v = \text{Re}(e^{i\theta_v} \prod_j A_{c_j}) = \cos \theta_v \prod_j A_{c_j}$ because the A 's are real. At the root, $F_r = e^{i\Theta}$ and $R_r = \prod_{v \neq r} \cos \theta_v$ with $\Theta = \sum_{v \neq r} \theta_v$, and $P_r = I$ gives

$$E_T^P = |e^{i\Theta} - \prod \cos \theta_v| = |1 - \prod w(\theta_v)|,$$

using $|1 - \bar{z}| = |1 - z|$. $\square$

Corollary 8.2. $C_T^P(\eta) \geq C_k(\eta)$.

Proof. In Theorem 8.1 take $\theta_v \equiv \theta_* \in \arg \max_{0 \leq \theta \leq \arcsin \eta} |1 - w(\theta)^{k-1}|$. For $k = 1$ both sides are 0. $\square$

Remark (The asymptotic witness). An earlier construction multiplied the phases of leaf slots as well and read an imaginary part, which gives $|\sin((k-1)\theta)|$. As written, it violates leaf-slot closure; routing leaves through $\text{Re}(\cdot)$ repairs it. The repaired family proves $\lim_{\eta \downarrow 0} C_T^P(\eta) = k-1$, but for $k \geq 3$ it does not attain $C_k(\eta)$. Indeed,

$$|1 - w(\theta)^n|^2 = \sin^2(n\theta) + (\cos^n \theta - \cos(n\theta))^2.$$

9. PROOF OF THEOREM R AND SCOPE

Proof of Theorem R. Theorem 7.1 and Corollary 8.2. □

Degenerate cases.

- $\eta = 0$ is outside the class.
- $\eta = 1$ with some $\theta = \pi/2$, or some $\rho_i = 0$: Lemma 4.2 and Lemma 6.3.
- $R_v = 0$ or $\tilde{R} = 0$: Lemma 4.2.
- $P_v \in \{0, I\}$: no division by $\|P_v g\|$ occurs.
- $\|F_v\| < 1$: allowed by Lemma 2.2.
- Angle labels above π : the truncation $\min(\cdot, \pi)$ and the case $\Theta > \pi$ in Lemma 6.3.
- Collinear or one-dimensional factors: the reductions in Theorem 3.1.
- Arity 0 and arbitrary arity: Lemma 4.2 and Theorem 3.1.
- Mixed leaf and internal children: (N2) for the upper bound, and the Re-gates for the witness.
- $k = 1$: Theorem 7.1 and Corollary 8.2.

Scope.

- *Complex spaces.* The upper bound plausibly transfers by realification; this is not claimed here. The witness has complex-bilinear norm $\sqrt{2}$, and the complex lower bound is open.
- *Directed acyclic graphs.* Proposition 5.1 fails as stated.
- *Other variants.* Shared laws, fixed ranks and oblique projectors are outside PMT-A.

Computational controls (not premises).

- Random falsification of Lemma 4.2: 344 000 node cases at arities 1–3, plus adversarial optimization. Every flagged case was a discretization artifact of the θ -grid.
- Cutting-plane LP evaluation of Theorem 3.1 for $m = 2, 3, 4$: the upper estimate is within $1.6 \cdot 10^{-7}$ of the bound.
- 14 000 degenerate node cases at the proof angle: largest violation $6.7 \cdot 10^{-16}$.
- The witness on 4000 random trees ($k \leq 10$, arity ≤ 4): attainment error $\leq 3.6 \cdot 10^{-15}$.
- The Diagonal Lemma for $n \leq 12$: no excess beyond grid resolution.

10. CONSEQUENCES

Write $n = k - 1$ and $s = \sin^2 \theta$. Then $|1 - w(\theta)^n|^2 = E_k(s)$, where

$$E_k(s) := 1 + (1 - s)^n - 2(1 - s)^{n/2} \cos(n\theta)$$

is a polynomial in s , because $\cos^n \theta \cos(n\theta)$ is an even polynomial in $\cos \theta$. Suppose E_k has a unique critical point $s_c(k)$ in $(0, \sin^2(\pi/n))$. Then

$$C_k(\eta) = \begin{cases} \sqrt{E_k(\eta^2)}/\eta, & \eta^2 \leq s_c(k), \\ \sqrt{E_k(s_c(k))}/\eta, & \eta^2 \geq s_c(k). \end{cases}$$

There are two regimes: below the critical leakage the extremizer uses the whole budget, and above it the absolute error saturates. Uniqueness of the critical point was verified in exact arithmetic for $3 \leq k \leq 12$.

k	$E_k(s)$	$s_c(k)$	$\max_\eta \eta C_k(\eta)$	a_k
2	s	1	1	0
3	$s(4 - 3s)$	$2/3$	$2/\sqrt{3}$	$3/4$
4	$s(9 - 15s + 7s^2)$	$3/7$	$9/7$	$5/2$
5	$s(4 - 3s)(4 - 8s + 5s^2)$	$\frac{3}{5} - \frac{\sqrt{21}}{15}$	1.38090...	$11/2$
6	$s(25 - 100s + 160s^2 - 115s^3 + 31s^4)$	$\frac{15 - \sqrt{70}}{31}$	1.45299...	10
7	$s(4 - 3s)(1 - 3s + 3s^2)(9 - 15s + 7s^2)$	cubic root	1.50959...	$65/4$

Proposition 10.1 (Second-order coefficient). *As $\eta \downarrow 0$,*

$$C_k(\eta) = (k-1) - a_k \eta^2 + O(\eta^4), \quad a_k = \frac{(k-1)(k-2)(k+6)}{24}.$$

Proof. For small η the maximum is attained at $\theta = \arcsin \eta$, since $|1 - w^n|$ is increasing near 0. We have $\log w(\theta) = i\theta - \theta^2/2 + O(\theta^4)$, so $X := n \log w = in\theta - n\theta^2/2 + O(\theta^4)$ and

$$|1 - e^X|^2 = n^2 \theta^2 + \left(\frac{n^2}{4} - \frac{n^3}{2} - \frac{n^4}{12}\right) \theta^4 + O(\theta^6).$$

Hence $|1 - w^n| = n\theta \left(1 + \left(\frac{1}{8} - \frac{n}{4} - \frac{n^2}{24}\right) \theta^2\right) + O(\theta^5)$. Substituting $\theta = \eta + \eta^3/6 + O(\eta^5)$ and dividing by η gives $n - \frac{n(n-1)(n+7)}{24} \eta^2 + O(\eta^4)$. $\square$

The coefficient grows like $k^3/24$. The sequence $3/4, 5/2$ at $k = 3, 4$ does *not* determine it: the quadratic $(k-2)(2k-3)/4$ fits both values but predicts $a_5 = 21/4$ instead of $11/2$.

Proposition 10.2 (Large k). *Let $G_k := \max_{\eta \in (0,1]} \eta C_k(\eta)$ and let $\eta_c(k)$ be the leakage at which the maximum over θ is first attained. Then $1 + \cos^{k-1}(\frac{\pi}{k-1}) \leq G_k < 2$, so $G_k \rightarrow 2$, and $\eta_c(k) \leq \sin(\frac{\pi}{k-1}) \rightarrow 0$.*

Proof. The proof of Lemma 6.3 shows that the maximum of $|1 - w^n|$ over $[0, \pi/2]$ is attained in $[0, \pi/n]$. At $\theta = \pi/n$ we have $w^n = -\cos^n(\pi/n)$. Moreover $|1 - w^n| \leq 1 + \cos^n \theta < 2$ for $\theta > 0$. Finally $\cos^n(\pi/n) = \exp(-\pi^2/(2n) + O(n^{-3})) \rightarrow 1$. $\square$

Numerically, $G_k = 2 - \pi^2/(2k) + o(1/k)$ and $k \eta_c(k) \rightarrow \pi$. Neither monotonicity in k nor these asymptotic constants are proved here.

Remark (Why the Löwner order O2). A weaker candidate order compares the rotated distances $|a - \lambda \zeta e^{i\varphi}|$ and $|1 - \lambda z e^{i\varphi}|$ for all $\lambda \in [0, 1]$ and $\varphi \in [0, \pi]$. It contains strictly more states: in random sampling about 7% of the states it dominates are not O2-dominated. It is not needed. What makes the induction close is that O2 allows the dilation angle to be chosen *smaller* than the accumulated label (Lemma 2.3). An earlier attempt that forced the angle to equal the label required case-by-case repairs and did not extend beyond four internal nodes.

QUESTIONS FOR THE REVIEWER

Question 1. Is Theorem 3.1 correct for arbitrary finite arity? Is it known, for instance in the theory of projective tensor norms, contractive dilations, or scaled relative graphs?

Question 2. Does the contraction and dilation argument in Lemma 4.2 correctly carry the raw pair (f, g) to a dominating chain pair at angle $\tilde{\psi} + \theta_v$? In particular, is Lemma 2.3 applied under its exact hypotheses?

Question 3. Does any step use closure beyond the trajectory form (TC)?

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